

Mohammad Refat

 [MohammadRefat23](#) |  [LinkedIn](#) |  [Website](#) |  moref17@gmail.com |

RESEARCH INTERESTS

Keywords: Computational astrophysics, Time-domain astronomy, Light-curve inversion, Stellar surface mapping, Brown dwarf and exoplanet atmospheres, Bayesian inference, Scientific computing

Executive Summary: I am a computational astrophysicist interested in developing statistical and computational methods for extracting physical insight from astronomical observations. My research combines time-series analysis, inverse problems, and Bayesian inference to study stellar surface features, brown dwarf and exoplanet atmospheres, and other time-domain phenomena. More broadly, I am interested in applying scientific computing and probabilistic modeling to understand complex astrophysical systems.

TEACHING

Adjunct Lecturer <i>Baruch College</i>	Aug 2025 – May 2026
– Introductory Physics Lecture (PHYS 2002)	Fall 2025, Spring 2026
– Introductory Physics Laboratory (PHYS 2003)	Fall 2025
– Physics II Laboratory (PHYS 3001)	Spring 2026

EDUCATION

City University of New York (CUNY) Graduate Center M.S. in Astrophysics Thesis: <i>Starspot Inference Using Light Curve Inversion Techniques</i>	2023–2025
CUNY Baccalaureate for Unique and Interdisciplinary Studies B.S. in Computational Astrophysics, <i>cum laude</i>	2019–2022
Bernard M. Baruch College	2017–2019

RESEARCH EXPERIENCE

Starspot Inference with Light Curve Inversion Techniques <i>American Museum of Natural History</i> Advisor: Dr. Lucy Lu	Aug 2023 – Sep 2025
– Investigated the feasibility and limitations of reconstructing stellar surface features from rotational photometric variability.	
– Applied light-curve inversion, time-series analysis, and Bayesian inference to study non-unique starspot configurations and model degeneracies.	
– Developed Python-based computational workflows for simulation, model fitting, and evaluation as part of an M.S. thesis on starspot inference.	
Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres <i>American Museum of Natural History</i> Advisor: Dr. Johanna Vos	May 2021 – Aug 2023
– Investigated atmospheric variability in brown dwarfs and giant exoplanets using rotational photometric	

observations.

- Applied time-series analysis and statistical modeling to characterize cloud-driven variability and infer atmospheric structure.
- Developed Python workflows for processing observational datasets and analyzing photometric light curves.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2 May 2020 – May 2021

Sloan Digital Sky Survey Faculty and Student Team

Advisor: Dr. Allyson Sheffield

- Investigated the chemical and dynamical properties of candidate stellar members of the Jhelum stellar stream using APOGEE-2 spectroscopy.
- Applied statistical analysis to identify stream members and characterize their chemical abundance patterns.
- Contributed to a refereed publication in *The Astrophysical Journal*.

Looking at Star Formation Through Chemistry

Aug 2018 – May 2020

American Museum of Natural History

Advisor: Dr. Allyson Sheffield

- Investigated stellar populations through spectroscopic measurements of chemical abundances.
- Processed and normalized echelle spectra and measured equivalent widths of spectral lines.
- Applied quantitative spectroscopic analysis techniques to characterize stellar chemical compositions.

Two-Point Statistics in the Star Forming ISM

Jun 2018 – Aug 2018

Flatiron Institute, Center for Computational Astrophysics

Advisor: Dr. Chang-Goo Kim

- Investigated statistical relationships between metallicity, gas dynamics, and star formation in the interstellar medium.
- Applied two-point statistical methods to analyze correlations within numerical simulation data.

PUBLICATIONS

Sheffield, Allyson A., Aidan Z. Subrahimovic, **Mohammad Refat**, et al. (May 2021). “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”. In: *The Astrophysical Journal* 913.1, p. 39. ISSN: 1538-4357. DOI: [10.3847/1538-4357/abee93](https://doi.org/10.3847/1538-4357/abee93). URL: <http://dx.doi.org/10.3847/1538-4357/abee93>.

Refat, Mohammad (2025). “Starspot Inference Using Light Curve Inversion Techniques”. Master’s thesis. CUNY Graduate Center. URL: https://academicworks.cuny.edu/gc_etds/6480/.

ACADEMIC PRESENTATIONS

Talks

“Starspot Inference with Light Curve Inversion Techniques” 2025
CUNY Masters Graduation 2025

Flatiron Institute, Center for Computational Astrophysics

“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” 2022
240th American Astronomical Society Meeting
American Astronomical Society

“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics / City University of New York / American Museum of Natural History Symposium</i> Flatiron Institute, Center for Computational Astrophysics	2021
“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” <i>American Museum of Natural History REU Symposium</i> American Museum of Natural History	2020
“Where Were Stars Born in the Milky Way?” <i>American Museum of Natural History REU Symposium</i> American Museum of Natural History	2019
“Two-Point Statistics in the Star Forming ISM” <i>Center for Computational Astrophysics REU Symposium</i> Flatiron Institute, Center for Computational Astrophysics	2018

Posters

“Starspot Inference with Light Curve Inversion Techniques” <i>245th American Astronomical Society Meeting</i> American Astronomical Society	2025
“Star Spots with Starry” <i>22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun</i> University of California San Diego	2024
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics Gotham Fest 2023</i> Flatiron Institute, Center for Computational Astrophysics	2023
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>241st American Astronomical Society Meeting</i> American Astronomical Society	2023
“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” <i>237th American Astronomical Society Meeting</i> American Astronomical Society	2021

TECHNICAL SKILLS

Programming	Python, MATLAB, Bash
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Jupyter Notebook, Astropy, Lightkurve, Astroquery
Computational Methods	Time-Series Analysis, Statistical Modeling, Inverse Problems, Bayesian Inference, Computational Modeling, Spectral Analysis
Software	Git, Linux, L ^A T _E X, Scientific Writing